



Advanced Computer Systems Engineering Laboratory (ENCS5150)

Android Course Project - Smart Activity/Travel Management Application

First Semester 2026

1. Overview

Submission Information: This is a group-based project for **2 students** and is worth **100 marks**. The submission deadline and discussion date are **to be announced**.

It is very important for students to carefully read the [GitHub Repository Requirements](#) before starting the project. In this project, students are required to design and implement an Android application using Java. The application should help users manage activities, events, reservations, favorites, and personal profiles in a modern and user-friendly environment. Students are encouraged to create modern UI designs and use their creativity while respecting the required functionalities. The project focuses on applying Android development concepts learned during the course.

2. Application Theme Selection

To reduce similarity between projects and encourage creativity, each group must select the application theme according to the last digit of the summation of both students' university IDs.

Last Digit	Required Theme
< 5	Right Theme
>= 5	Left them

Each group must fully design their application according to the assigned theme.



3. Application Type Selection

To reduce similarity between projects and encourage creativity, each group must select the application type according to the last digit of the smallest students' university IDs.

Last Digit	Required Theme
even	Smart University Events App
odd	Travel Planner App

Each group must fully design their application according to the assigned type.

4. General Requirements

4.1. Splash Screen & Introduction Layout

Create an animated Splash Screen that displays the application logo for 2–3 seconds. After the animation finishes, display an Introduction Layout containing: Application title (you are selected it at [3. Application Type Selection](#)), Short description about the application, “Connect” button

4.1.1. Connect Button Behavior

The button should connect to a RESTful API to fetch external data (events/trips) related to the application type, each API must contain **at least 10 objects (events/trips)** in the array.

4.1.1.1. Smart University Events App

```
[
  {
    "id": 1,
    "title": "AI Workshop",
    "description": "Introduction to AI",
    "category": "Technology",
    "date": "2026-06-10",
    "time": "10:00 AM",
    "location": "Engineering Hall",
    "seats": 80,
    "image": "https://example.com/ai.jpg"
  }
]
```

4.1.1.2. Travel Planner App

```
[
  {
    "id": 101,
    "destination": "Istanbul",
    "country": "Turkey",
    "duration_days": 5,
    "price": 750,
    "rating": 4.7,
    "description": "Explore historical",
    "image": "https://exp.com /istanbul.jpg"
  }
]
```

4.1.2. Requirements

- If the connection succeeds → navigate to Login/Register screen, and store the API result in database.
- If the connection fails → display an error Toast message and remain on the same screen.

4.2. Login & Registration

4.2.1. Login Screen

Should contain: Email field, Password field, Remember Me checkbox, Login button, Sign-Up button

4.2.1.1. Requirements

- Validate user inputs before login.
- “Remember Me” must save the email using Shared Preferences (Password must be encrypted).
- If login succeeds → navigate to Home Layout.
- If login fails → display an error message.

4.2.2. Registration Screen

The registration screen must include the following fields:

- Email (must follow email format)
- First Name (minimum 3 characters)
- Last Name (minimum 3 characters)
- Password
- Confirm Password
- Gender (Spinner)
- Major or Category (Spinner)
- Phone Number

4.2.2.1. Password Conditions

- Contain at least 1 letter
- Contain at least 1 number
- Be at least 6 characters long

4.2.2.2. Additional Requirements

- Confirm Password must match Password.
- All inputs must be validated.
- Password should be stored securely (stores the password in encrypted way).
- After successful registration → redirect to Login Screen.

4.3. Main Home Layout

The application must use a Navigation Drawer.

The Home page should contain:

- Application overview
- Theme description
- Attractive UI design

4.3.1 Navigation Drawer Items

- Home
- Events / Trips (Depends on your selection at [3. Application Type Selection](#))
- My Reservations
- Favorites
- Special Section
- Profile Management
- Contact Us
- Logout

4.4. Events/Trips Section

This section represents the main functionality of the application, each event/trip should include all information available at the Restful API

4.4.1 Requirements

- Display data using RecyclerView.
- Use Fragments to display details.
- Add Search functionality.
- Add Filtering functionality.
- Add “Favorite” button.
- Add Reservation/Join button.

4.4.2 Reservation Form

When the user clicks the reservation/join button, open a form containing:

- Quantity or participation count
- Reservation type
- Confirmation button

The reservation must be stored in the application database.

4.5. My Reservations Section

This section should display all previous and current reservations; each reservation must include:

- event/trip name
- Reservation date
- Status
- Additional information

4.6. Favorites Section

This section should display all favorite items added by the user, the user should be able to:

- Remove items from favorites
- Open item details
- Make reservations directly

4.7. Special Section

Each application type must include a **Special Section** related to its idea.

Examples:

- **Smart University Events App:** Featured Events, Popular Events, Upcoming Important Events.
- **Travel Planner App:** Travel Offers, Popular Destinations, Recommended Trips.

Students are free to decide how to define the special section in their application. For example, they can mark an item as special based on category, rating, date, discount, popularity.

The Special Section must support:

- Displaying special items
- Opening item details
- Adding items to favorites

4.8. Profile Management

The profile screen should display user information.

Users must be able to:

- View profile information
- Update first name
- Update last name
- Update password
- Update phone number
- Add profile picture

4.8.1 Requirements

- Validation rules must still apply.
- Password conditions must be respected.

4.9. Contact Us Layout

Create a Contact Us screen containing three buttons:

- Call Us: Open the phone application with a predefined number.
- Locate Us: Open Google Maps with a predefined location.
- Email Us: Open Gmail with a predefined email address.

4.10. Logout

The logout functionality should:

- End the current session
- Return the user to the Login Screen

4.11. Admin Home Layout

The application must include a pre-created Admin account:

- Email: admin@admin.com
- Password: Admin123!

The admin should have a separate Navigation Drawer.

4.11.1 Admin Functionalities

- Add new admin
- View users
- Delete users
- Add new events/trips
- Edit events/trips
- Delete events/trips
- View reservations
- Logout

5. Technical Requirements

The application must demonstrate the use of the following Android concepts:

- Android Layouts (Static & Dynamic)
- Navigation Drawer
- RecyclerView
- Fragments
- Intents
- Notifications (Toast or Push Notifications)
- Shared Preferences
- Database Management
- RESTful API Integration
- Animations
- Input Validation
- Exception Handling

6. Animation Requirements

The application must include at least two animations.

Examples:

- Fade Animation
- Bounce Animation
- Slide Animation
- Button Scale Animation
- Splash Screen Animation

7. UI & Design Requirements

Students are encouraged to:

- Create attractive interfaces
- Use proper colors and icons
- Design responsive layouts
- Follow modern Android design principles

Creativity and UI quality will be considered during grading.

Discussion Session

A discussion session will be scheduled after submission; both team members must attend failure to attend the discussion session will result in a grade of zero.

Students may be asked to:

- Explain code, modifying features, add small functionalities
- Explain database structure, explain API integration

GitHub Repository Requirements

For this project, each group must use a **PRIVATE** GitHub repository to manage and track development work throughout the semester.

- Only one student per group should create a private repository. The second student must be added as a contributor/collaborator.
- Students must push their work feature by feature as the project develops.
- The repository should show regular progress, with at least 30-40 commits throughout the project.
- Uploading the major project files as one large block at the end will result in a grade of zero.
- The expected workflow is to implement one feature in a commit, then continue improving it over time by fixing bugs or making small updates in later commits.
- Using separate branches and pull requests or merge requests is optional, but strongly recommended as good development practice.

Notes

- The name of the application must be: **FirstStudentID_SecondStudentID_CourseProject**
- Use **Pixel 3a XL device with API Level 28 (Graphic = Software)**, the **minimum SDK** should be **API 28**.
- **No late submissions whatsoever**. If the deadline is about to arrive, submit your work as it is.
- Creativity and additional features may improve the final evaluation.
- Projects must be implemented by the students themselves.
- Copying code between groups is strictly prohibited.
- Similar projects may receive zero marks, Cheating or using LLM Models will result in **0 marks**.
- Students are allowed to use online resources for learning purposes only.

Submission Requirements

Send both the APK file and the ZIP file as a reply to the project submission message.

1. **Project.zip file** (Size in KB)
 - From Android Studio: File → Export → Export to Zip File
2. **app-debug.apk file** (Size in MB)
 - From Android Studio: Build → Build Bundle(s) / APK(s) → Build APK(s)
 - You will find the APK file under: app\build\outputs\apk\debug\app-debug.apk
3. **Private GitHub repository link** must be included in the message submission.